

Drilling engineering — pore pressure, hydraulics & well control

Closed-form, license-free drilling-geomechanics and hydraulics engines: multi-method pore-pressure prediction with explicit P10/P50/P90 uncertainty and a communicated safe mud-weight window, swab/surge tripping limits, annular hydraulics and ECD, ROP models, and a real-time dysfunction / kick detector. Every method is a published closed form with unit tests pinned to hand calcs and standard limiting cases — distinct from the casing-design explorer under Well construction, which covers tubular strength.

P10/P50/P90

pore-pressure uncertainty

8

Bourgoyne-Young ROP parameters

WHAT YOU GET

- Multi-method pore pressure live: Eaton and Bowers gradients diverging down an overpressured well, governing minimum mud weight and the kick/loss mud-weight window
- Swab / surge from Burkhardt's clinging constant through Bingham annular friction, solved for the max trip speed that keeps BHP inside the window
- Generalised annular hydraulics per API RP 13D (velocity, frictional loss, ECD, cuttings transport) for arbitrary pipe-in-hole geometry
- Bourgoyne-Young 8-parameter and Warren power-law ROP models
- Real-time dysfunction & kick detector over surface-parameter windows: stick-slip, washout, bit-balling, instability and kick with severity and mitigation